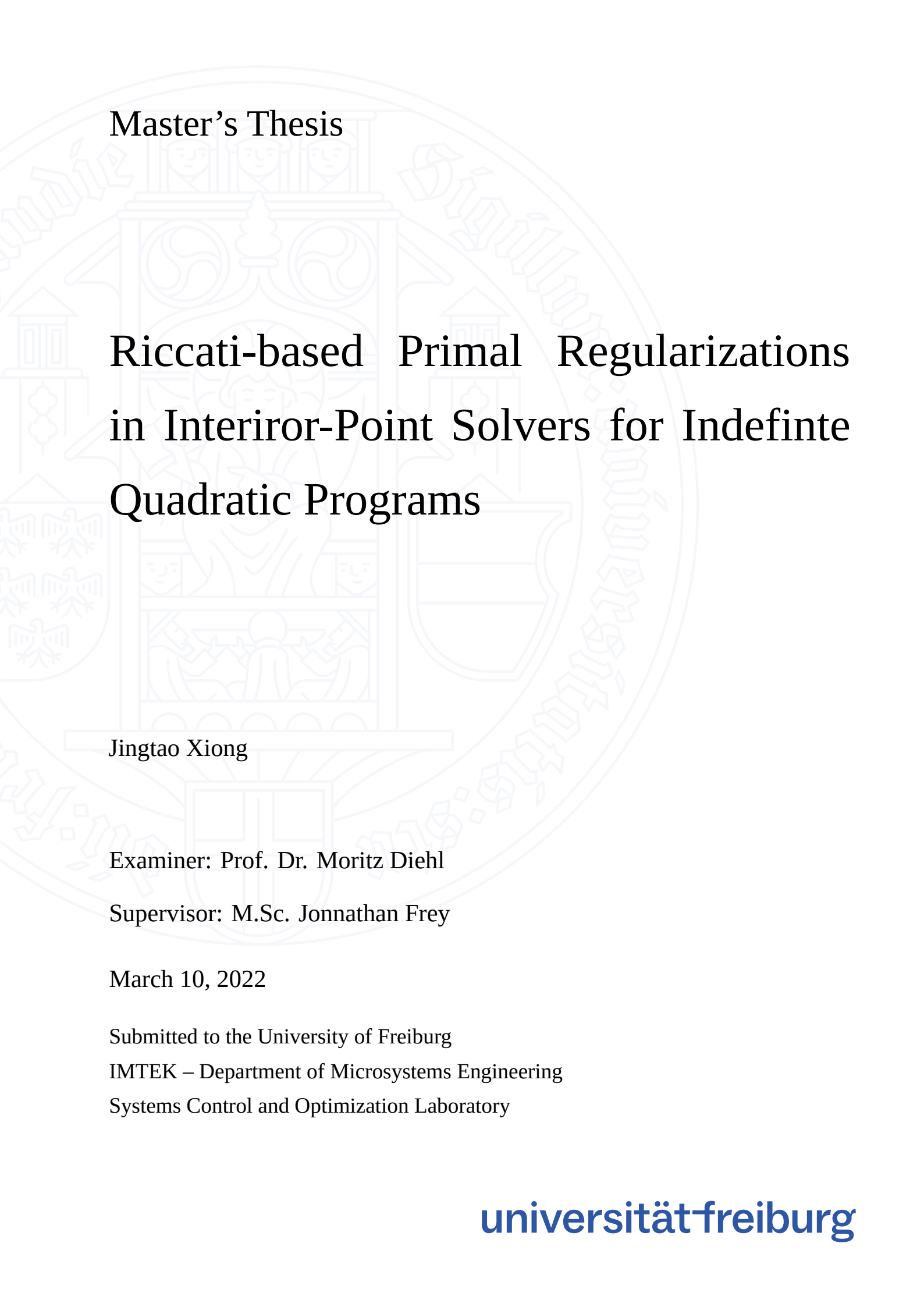




Master's Thesis

Riccati-based Primal Regularizations in Interior-Point Solvers for Indefinite Quadratic Programs

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Chapter 1

Primal–Dual Interior Point Solver in Optimal Control Problems

This chapter provides background knowledge to understand the concepts in the thesis.

1.1 Primal–Dual Interior Point Method

This section will introduce the primal-dual interior point method (IPM) for solving quadratic programming (QP) problems with equality and inequality constraints [1], which is the core algorithm for solving the optimal control problems in this thesis.

Consider a QP with equality $Fy = b$ and inequality constraints $Gy \leq d$. By introducing slack variables t , we have the following formulation:

$$\begin{aligned} & \underset{y, t}{\text{minimize}} && \frac{1}{2}y^\top Hy + g^\top y \\ & \text{subject to} && \begin{aligned} & Fy = b, \\ & Gy + t = d, \\ & t \geq 0, \end{aligned} \end{aligned} \tag{1.1}$$

we can further obtain the lagrangian function of the QP problem (1.1) as

$$\mathcal{L}(y, t, \pi, \lambda) = \frac{1}{2}y^\top Hy + g^\top y + \pi^\top (Fy - b) + \lambda^\top (Gy + t - d), \tag{1.2}$$

where π and $\lambda \geq 0$ are the Lagrange multipliers associated with the equality and inequality constraints, respectively. Then the KKT conditions of the QP

problem (1.1) are given by

$$Hy + g + F^\top \pi + G^\top \lambda = 0, \quad (1.3a)$$

$$Fy - b = 0, \quad (1.3b)$$

$$Gy + t - d = 0, \quad (1.3c)$$

$$\lambda_j t_j = 0, \quad j = 1, \dots, n_c, \quad (1.3d)$$

$$\lambda \geq 0, \quad t \geq 0. \quad (1.3e)$$

where n_c is the number of inequality constraints. The definiteness of the Hessian matrix H distinguishes the QP problem (1.1) as convex or nonconvex. In primal-dual IPM, a homotopy parameter τ , also called as barrier parameter, is applied to the only nonlinear equation (1.3d) for relaxing the complementarity, the equation (1.3d) can be rewritten as:

$$\lambda_j t_j = \tau, \quad j = 1, \dots, n_c, \quad (1.4)$$

the parameter τ is shrunk toward zero as the iterations approach the solution with certain strategy, and size step control is applied to ensure the satisfaction of inequality (1.3e). The corresponding update strategy and step size control will be discussed in the following sections.

In order to solve the nonlinear system of equations (1.3), the Newton's method is applied to the linearized system at each iteration i to obtain the step $(\Delta y, \Delta \pi, \Delta \lambda, \Delta t)$, which takes the form of:

$$\begin{bmatrix} H & F^\top & G^\top & 0 \\ F & 0 & 0 & 0 \\ G & 0 & 0 & I \\ 0 & 0 & T_i & \Lambda_i \end{bmatrix} \begin{bmatrix} \Delta y \\ \Delta \pi \\ \Delta \lambda \\ \Delta t \end{bmatrix} = - \begin{bmatrix} r_{\text{stat}}^i \\ r_{\text{eq}}^i \\ r_{\text{ineq}}^i \\ r_{\text{comp}}^i \end{bmatrix}, \quad (1.5)$$

With the diagonal matrices $T_i = \text{diag}(t_i)$, $\Lambda_i = \text{diag}(\lambda_i)$, and the perturbed KKT residuals follow as:

$$\begin{bmatrix} r_{\text{stat}}^i \\ r_{\text{eq}}^i \\ r_{\text{ineq}}^i \\ r_{\text{comp}}^i \end{bmatrix} := \begin{bmatrix} Hy_i + g + F^\top \pi_i + G^\top \lambda_i \\ Fy_i - b \\ Gy_i + t_i - d \\ T_i \Lambda_i \mathbf{1} - \tau_i \mathbf{1} \end{bmatrix}, \quad (1.6)$$

Based on the structure of the KKT system (1.5), Schur complement reduction can be applied to reduce the system (1.5) to a saddle point problem, the symmetric system can be presented as:

$$\begin{bmatrix} \tilde{H}^i & F^\top \\ F & 0 \end{bmatrix} \begin{bmatrix} \Delta y \\ \Delta \pi \end{bmatrix} = - \begin{bmatrix} \tilde{r}_{\text{stat}}^i \\ r_{\text{eq}}^i \end{bmatrix}, \quad (1.7)$$

with $\tilde{H}^i = H + G^\top T_i^{-1} \Lambda_i G \succeq H$ and $\tilde{r}_{\text{stat}}^i = r_{\text{stat}}^i + G^\top T_i^{-1} (\Lambda_i r_{\text{ineq}}^i - r_{\text{comp}}^i)$. Notably, the additional term in the Hessian $G^\top T_i^{-1} \Lambda_i G$ can be regarded as the curve contribution from active constraints, which is positive semidefinite. This term can already be interpreted as a kind of regularization in the problems.

In the (1.7), assuming F has full row rank, the null space method can be sequentially applied to reduce the system. With Y, Z spanning the range and null space of F , respectively, the primal step can be decomposed as $\Delta y = Y \Delta y_y + Z \Delta y_z$. This yields the Schur-complement based reduced-Hessian system:

$$\tilde{M}^i \Delta y_z = - (Z^\top \tilde{r}_{\text{stat}}^i + Z^\top \tilde{H}^i Y \Delta y_y), \quad (1.8)$$

where the equality Jacobian F is eliminated by $Z^\top$ (with $Z^\top F^\top = 0$), and feasibility fixes the range-space component $\Delta y_y = -(FY)^{-1} r_{\text{eq}}^i$, the complemented reduced Hessian $\tilde{M}^i = Z^\top \tilde{H}^i Z$ and the right-hand side $Z^\top \tilde{r}_{\text{stat}}^i + Z^\top \tilde{H}^i Y \Delta y_y$. This formulation exposes the connection between the reduced Hessian $\tilde{M}^i$ and the original Hessian $\tilde{H}$.

1.2 Efficient Solution with Riccati Recursion

Within the context of optimal control problems, the KKT system (1.7) exhibits a tridiagonal structure due to the temporal nature of the problem, as:

$$\begin{bmatrix} \ddots & & & & & \\ & 0 & -I & & & \\ & -I \begin{pmatrix} \tilde{Q}_k & \tilde{S}_k^\top \end{pmatrix} A_k^\top & & & & \\ & 0 \begin{pmatrix} \tilde{S}_k & \tilde{R}_k \end{pmatrix} B_k^\top & & & & \\ & 0 & A_k & B_k & 0 & -I \\ & & & & -I & \ddots \end{bmatrix} \begin{bmatrix} \vdots \\ \Delta \pi_k \\ \Delta x_k \\ \Delta u_k \\ \Delta \pi_{k+1} \\ \vdots \end{bmatrix} = - \begin{bmatrix} \vdots \\ r_{\text{eq}}^{k-1} \\ \tilde{r}_{\text{stat},x}^k \\ \tilde{r}_{\text{stat},u}^k \\ r_{\text{eq}}^k \\ \vdots \end{bmatrix}, \quad (1.9)$$

where A_k and B_k denote the Jacobians of the equality constraints with respect to the state and control variables at time step k , respectively. And the block matrices $\tilde{Q}_k$, $\tilde{R}_k$, and $\tilde{S}_k$ are defined as:

$$\begin{aligned}\tilde{Q}_k &= Q_k + (G_x^k)^\top \Lambda_k T_k^{-1} G_x^k, \\ \tilde{R}_k &= R_k + (G_u^k)^\top \Lambda_k T_k^{-1} G_u^k, \\ \tilde{S}_k &= S_k + (G_x^k)^\top \Lambda_k T_k^{-1} G_u^k,\end{aligned}$$

where Q_k , R_k , and S_k are the original block components constituting the Hessian matrix H_k , stands for state, control, and cross terms, respectively. G_x^k and G_u^k denote the corresponding Jacobians for the inequality constraints and possible box constraints on state and control.

This tridiagonal structure allows for efficient solution of the KKT system using Riccati recursion [2], which is consisted of a backward and forward pass. The backward pass computes the feedback gains K and cost-to-go matrices P , while the forward pass computes the optimal state and control trajectories, the algorithmic details of the Riccati recursion can be summarized as follows:

Algorithm 1 Riccati Recursion for Solving OCP Structure QP

```
1: Initialize:  $P_N \leftarrow \tilde{Q}_N$ ,  $p_N \leftarrow \tilde{r}_{\text{stat},x}^N$ 

  // Backward recursion
2: for  $k = N - 1$  down to 0 do
3:    $\tilde{M} \leftarrow \tilde{R}_k + B_k^\top P_{k+1} B_k$ 
4:    $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P_{k+1} A_k$ 
5:    $K_k \leftarrow -\tilde{M}^{-1} \Gamma_k$ 
6:    $k_k \leftarrow -\tilde{M}^{-1} \left( \tilde{r}_{\text{stat},u}^k + B_k^\top (P_{k+1} r_{\text{eq}}^k + p_{k+1}) \right)$ 
7:    $P_k \leftarrow \tilde{Q}_k + A_k^\top P_{k+1} A_k - \Gamma_k^\top \tilde{M}^{-1} \Gamma_k$ 
8:    $p_k \leftarrow \tilde{r}_{\text{stat},x}^k + A_k^\top (P_{k+1} r_{\text{eq}}^k + p_{k+1}) + \Gamma_k^\top k_k$ 
9: end for

  // Forward recursion
10: for  $k = 0$  to  $N - 1$  do
11:    $\Delta u_k \leftarrow K_k \Delta x_k + k_k$ 
12:    $\Delta x_{k+1} \leftarrow A_k \Delta x_k + B_k \Delta u_k + r_{\text{eq}}^k$ 
13:    $\Delta \pi_{k+1} \leftarrow P_{k+1} \Delta x_{k+1} + p_{k+1}$ 
14:    $\Delta t_k \leftarrow -(G_x^k \Delta x_k + G_u^k \Delta u_k + r_{\text{ineq}}^k)$ 
15:    $\Delta \lambda_k \leftarrow T_k^{-1} (-r_{\text{comp}}^k - \Lambda_k \Delta t_k)$ 
16: end for
```

By applying the Riccati recursion, we implicitly solve the KKT system (1.8) without explicitly forming the reduced Hessian $\tilde{M}^i$, and the computational complexity is reduced from $\mathcal{O}(N^3)$ to $\mathcal{O}(N)$, where N is the number of time steps in the optimal control problem.

1.3 Adaptive Barrier Update

Discuss some updating strategies for the barrier parameter τ in the primal-dual interior point method.

1.3.1 Mehrotra’s Predictor-Corrector

For interior-point method, the Mehrotra’s predictor-corrector scheme is usually considered as the default choice for fast convergence [3], which adaptively updates the barrier parameter τ and adds a second-order corrector for improving the direction, at essentially the cost of one extra back-solve. In the stratdety, the barrier parameter τ is defined as:

$$\tau_i = \sigma_i \mu_i, \quad \mu_i = \frac{t_i^\top \lambda_i}{m}, \quad (1.10)$$

where μ_i is the duality measure, m is the number of inequality constraints, and $\sigma_i \in [0, 1]$ is the centering parameter that controls the trade-off between centering and aggressive steps. And in each iteration, two solves of (1.5) that share the same left-hand side and differ only in the complementarity right-hand side are performed, which involves two steps: predictor step (affine scaling step), and corrector step.

Predictor Step. The strategy starts with computing a predictor step, the linear system (1.5) is solved with $\sigma_i = 0$ in the complementarity residual r_{comp}^i and no correction considered, i.e. the resulting complementarity residual is written as:

$$r_{\text{comp}}^{\text{aff}} = T_i \Lambda_i \mathbf{1} - \sigma_i \mu_i \mathbf{1} = T_i \Lambda_i \mathbf{1}, \quad (1.11)$$

to obtain the affine scaling direction $(\Delta y_{\text{aff}}, \Delta \pi_{\text{aff}}, \Delta \lambda_{\text{aff}}, \Delta t_{\text{aff}})$. The affine step is used to estimate the centering parameter σ_i for the next step.

Centering. After the predictor step is obtained, The centering parameter σ is updated based on the estimated duality measure μ_{aff} after the affine step, which is defined as:

$$\sigma_i = (\mu_{\text{aff}}/\mu_i)^3 \quad (1.12)$$

$$\mu_{\text{aff}} = \frac{(t_i + \alpha_{\text{aff}}^{\text{prim}} \Delta t_{\text{aff}})^\top (\lambda_i + \alpha_{\text{aff}}^{\text{dual}} \Delta \lambda_{\text{aff}})}{m}, \quad (1.13)$$

where the step size $\alpha_{\text{aff}}^{\text{prim}}$ and $\alpha_{\text{aff}}^{\text{dual}}$ are determined by ensuring that the updated slack variables and Lagrange multipliers remain positive, i.e.

$$\alpha_{\text{aff}}^{\text{prim}} = \min \left(1, \min_{j: \Delta t_{\text{aff},j} < 0} -\frac{t_{i,j}}{\Delta t_{\text{aff},j}} \right), \quad (1.14)$$

$$\alpha_{\text{aff}}^{\text{dual}} = \min \left(1, \min_{j: \Delta \lambda_{\text{aff},j} < 0} -\frac{\lambda_{i,j}}{\Delta \lambda_{\text{aff},j}} \right). \quad (1.15)$$

If $\mu_{\text{aff}} \ll \mu_i$, the affine step achieves a substantial reduction in the duality gap, which sharply gives $\sigma_i \rightarrow 0$ (near-Newton, aggressive), while a stalled step ($\mu_{\text{aff}} \approx \mu_i$) gives $\sigma_i \rightarrow 1$ (recentering toward the central path).

Corrector Step. With the updated centering parameter σ_i and calculated affine steps, the corrector step is computed by solving the linear system (1.5) again. In the corrector step, the complementarity residual is modified to include a second-order correction term that accounts for the nonlinear effects of the affine step, which is given by:

$$r_{\text{comp}}^{\text{cc}} = T_i \Lambda_i \mathbf{1} + \Delta T_{\text{aff}} \Delta \Lambda_{\text{aff}} \mathbf{1} - \sigma_i \mu_i \mathbf{1}, \quad (1.16)$$

The term $\Delta T_{\text{aff}} \Delta \Lambda_{\text{aff}} \mathbf{1}$ cancels the second-order error that the affine step leaves in $t_j \lambda_j$, and $-\sigma_i \mu_i \mathbf{1}$ supplies the centering.

This second solve returns the final direction $(\Delta y, \Delta \pi, \Delta \lambda, \Delta t)$ directly, a step size control will be followed to ensure the positivity of the slack variables and Lagrange multipliers.

In practice, the corrector step is usually applied conditionally, the mechanism is, therefore, called conditional Mehrotra's predictor-corrector step [4], where the corrector step is only applied if the affine step does not largely increase the estimated duality measure, i.e., $\mu_{\text{aff}} \leq 2\mu_i$.

1.3.2 Mehrotra’s Probing

However, Mehrotra’s predictor-corrector scheme is not always the best choice, particularly in the nonconvex case. Much as SQP methods form and use a local Hessian at each step, the nonconvex QP requires regularizing its possibly indefinite Hessian at every iterate to obtain a valid descent direction. The second-order correction in corrector step (1.16) may not be able to ensure a descent direction in the nonconvex case.

As mentioned in [4], the Mehrotra’s probing strategy can be applied alternatively to the predictor-corrector scheme, which taking a more conservative approach to update the barrier parameter τ by discarding the corrector step in predictor-corrector scheme but keeps the centering step. The corresponding complementarity residual is given by:

$$r_{\text{comp}}^{\text{probe}} = T_i \Lambda_i \mathbf{1} - \sigma_i \mu_i \mathbf{1}, \quad (1.17)$$

where the centering parameter σ_i is updated using (1.12) after the affine step, as in the predictor-corrector scheme.

1.4 Step Size Control

Discuss the step size control strategies in primal-dual interior point method, which is used to ensure the positivity of the slack variables and Lagrange multipliers.

1.4.1 Fraction-to-the-Boundary Rule

The fraction-to-the-boundary (FTB) rule is a common strategy for step size control in primal-dual interior point methods. The maximal step size $\alpha_{i,\max}^{\text{prim}}$ and $\alpha_{i,\max}^{\text{dual}}$ are computed without violating the positivity of the slack variables and Lagrange multipliers, respectively, as:

$$\alpha_{i,\max}^{\text{prim}} = \min_{j:\Delta t_j < 0} -\frac{t_{i,j}}{\Delta t_j}, \quad (1.18)$$

$$\alpha_{i,\max}^{\text{dual}} = \min_{j:\Delta \lambda_j < 0} -\frac{\lambda_{i,j}}{\Delta \lambda_j}. \quad (1.19)$$

Then, in practice, the actual step sizes are chosen as a fraction of the maximum step sizes, i.e.,

$$\alpha_i^{\text{prim}} = \min(1, \eta \alpha_{i,\max}^{\text{prim}}), \quad (1.20)$$

$$\alpha_i^{\text{dual}} = \min(1, \eta \alpha_{i,\max}^{\text{dual}}), \quad (1.21)$$

where $\eta \in (0, 1)$ is a parameter that controls the fraction of the maximum step size to be taken, typically set to a value close to 1 to ensure sufficient progress while maintaining feasibility.

1.4.2 Funnel Line Search with Backtracking

In nonconvex cases, the FTB rule may not be sufficient to guarantee convergence, as the search direction with FTB-controlled step size may not descend.

A funnel line search inspired by the funnel method [5] is implemented after the FTB rule applied to the step size, where the funnel is defined as a relaxation of the feasible set that allows a constraint violation up to a given upper bound $\theta_i > 0$.

Algorithm 2 Funnel line search

```

1: Input: iterate  $z$ , direction  $d$ , steps  $\alpha_p, \alpha_d$ ,  $f$ ,  $\theta = \|c\|$ ,  $\theta_{\max}$ ,  $\rho = \nabla f^\top d$ 
2: for  $ls = 1$  to  $\max\_ls$  do
3:   form candidate  $z^+(\alpha_p, \alpha_d)$ ; compute  $f^+$ ,  $\theta^+ = \|c^+\|$ 
4:   accepted  $\leftarrow$  false
5:   if  $\theta^+ \leq \theta_{\max}$  then ▷ in funnel
6:     if  $\alpha_p(-\rho) > \kappa_s \theta^2$  then ▷ switching
7:       if  $f^+ \leq f + \gamma \alpha_p \rho$  then
8:         accepted  $\leftarrow$  true ▷ f-step (Armijo)
9:       end if
10:    else if  $\theta^+ \leq \kappa_h \theta_{\max}$  then
11:       $\theta_{\max} \leftarrow (1 - \kappa) \theta^+ + \kappa \theta_{\max}$ 
12:      accepted  $\leftarrow$  true ▷ h-step
13:    end if
14:  end if
15:  if accepted then break
16:  else  $\alpha_p \leftarrow \beta \alpha_p$ 
17:  end if
18: end for
19: return  $\alpha_p, \alpha_d$ 

```

As summarized in Algorithm 2. At each trial of the funnel line search, a candidate iterate z^+ is formed from (α_p, α_d) and its objective f^+ and constraint violation $\theta^+ = \|c^+\|$ are evaluated. The iterate is *in the funnel* only if $\theta^+ \leq \theta_{\max}$, otherwise it is rejected and α_p is contracted. Inside the funnel, the *switching condition* decides the step type: if it holds, sufficient objective progress is expected and an *f-step* is taken, accepted when the Armijo condition is met; otherwise an *h-step* targets feasibility, accepted when $\theta^+ \leq \kappa_h \theta_{\max}$, which also tightens the funnel via $\theta_{\max} \leftarrow (1 - \kappa)\theta^+ + \kappa\theta_{\max}$. If neither is accepted, α_p is reduced and the search repeats.

Chapter 2

Riccati-based Primal Regularization

2.1 Riccati Recursion as explicit $L^\top DL$ decomposition

As mentioned in [6], the Riccati recursion can be interpreted as a block factorization of the KKT system 1.7. In this subsection, we further show the relationship between the Riccati recursion and the block $L^\top DL$ of reduced Hessian $\tilde{M}^i$.

On the one hand, the backward pass of the Riccati recursion can be concluded as follows:

Algorithm 3 Backward Pass of Riccati Recursion

```
1: Initialize:  $P_N \leftarrow \tilde{Q}_N$ ,  $p_N \leftarrow \tilde{r}_{\text{stat},x}^N$ 

2: for  $k = N - 1$  down to 0 do
3:    $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1} B_k$ 
4:    $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P_{k+1} A_k$ 
5:    $K_k \leftarrow -\tilde{M}_k^{-1} \Gamma_k$ 
6:    $k_k \leftarrow -\tilde{M}_k^{-1} \left( \tilde{r}_{\text{stat},u}^k + B_k^\top (P_{k+1} r_{\text{eq}}^k + p_{k+1}) \right)$ 
7:    $P_k \leftarrow \tilde{Q}_k + A_k^\top P_{k+1} A_k - \Gamma_k^\top K_k$ 
8:    $p_k \leftarrow \tilde{r}_{\text{stat},x}^k + A_k^\top (P_{k+1} r_{\text{eq}}^k + p_{k+1}) + \Gamma_k^\top k_k$ 
9: end for
```

Where the M_k is the reduced Hessian block obtained by Riccati recursion. The block $L^\top DL$ factorization of the reduced Hessian $\tilde{M}^i$ can be expressed

as:

$$\tilde{M}^i = L^\top \begin{bmatrix} \tilde{M}_0 & 0 & \cdots & 0 \\ 0 & \tilde{M}_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \tilde{M}_{N-1} \end{bmatrix} L$$

Consequently, the L is defined as:

$$L = \begin{bmatrix} I & 0 & \cdots & 0 \\ -K_1 B_0 & I & \cdots & 0 \\ -K_2 \Phi_{2,1} B_0 & -K_2 B_1 & \cdots & 0 \\ -K_3 \Phi_{3,1} B_0 & -K_3 \Phi_{3,2} B_1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ -K_{N-1} \Phi_{N-1,1} B_0 & -K_{N-1} \Phi_{N-1,2} B_1 & \cdots & I \end{bmatrix},$$

where $\Phi_{i,j}$ is defined as:

$$\Phi_{i,j} = \begin{cases} \prod_{k=j}^{i-1} A_k = A_{i-1} A_{i-2} \cdots A_j, & \text{if } i > j, \\ I, & \text{if } i = j. \end{cases}$$

2.2 Block Wise Primal Regularization

With this interpretation, we can directly apply the regularization to the reduced Hessian block $\tilde{M}_k^i$ during the Riccati recursion. When the Cholesky factorization of $\tilde{M}_k^i$ fails in practice, we do regularization to the reduced Hessian block $\tilde{M}_k^i$ by adding a regularization matrix block E_k^i , i.e. $\tilde{M}_k^i + E_k^i$. By the relationship, this can also guarantee the positive definiteness of $\tilde{M}^i$. The Backward pass of modified Riccati recursion can be concluded as follows:

Algorithm 4 Backward Pass of Modified Riccati Recursion

```

1: Initialize:  $P_N \leftarrow \tilde{Q}_N$ ,  $p_N \leftarrow \tilde{r}_{\text{stat},x}^N$ ,  $\lambda > 0$ 

2: for  $k = N - 1$  down to 0 do
3:    $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1} B_k$ 
4:    $L_k \leftarrow \text{Cholesky}(\tilde{M}_k)$ 
5:   if factorization fails then
6:      $\tilde{M}_k \leftarrow \tilde{M}_k + E_k$  ▷ Apply regularization
7:      $L_k \leftarrow \text{Cholesky}(\tilde{M}_k)$ 
8:   end if
9:    $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P_{k+1} A_k$ 

10:   $K_k \leftarrow -L_k^{-\top} L_k^{-1} \Gamma_k$ 
11:   $k_k \leftarrow -L_k^{-\top} L_k^{-1} \left( \tilde{r}_{\text{stat},u}^k + B_k^\top (P_{k+1} r_{\text{eq}}^k + p_{k+1}) \right)$ 

12:   $P_k \leftarrow \tilde{Q}_k + A_k^\top P_{k+1} A_k + \Gamma_k^\top K_k$ 
13:   $p_k \leftarrow \tilde{r}_{\text{stat},x}^k + A_k^\top (P_{k+1} r_{\text{eq}}^k + p_{k+1}) + \Gamma_k^\top k_k$ 
14: end for

```

2.3 Failing of Minimal Perturbation in Block Reduced Hessian

2.3.1 Minimal Example

In this subsection, we will discuss on the choice of regularization matrix block E_k^i in the stage regularization. We will show that the minimum perturbation strategy, i.e PROJECT[**verschuerenSparsityPreservingConvexification2017**], which is commonly the first choice for regularization, can cause condition number issue in the stage regularization.

Suppose at the stage k , the reduced Hessian block $\tilde{M}_k$ is not positive definite. Without loss of generality, we can assume $\tilde{M}_k$ has only one negative eigenvalue λ_j , and the corresponding eigenvector is v_j . The minimum perturbation strategy is applied as:

$$E_k = (-\lambda_j + \epsilon) v_j v_j^\top, \quad (2.1)$$

where the ϵ is a positive value to ensure the positive definiteness of the regularized reduced Hessian block.

By Sherman-Morrison formula, the inverse of the regularized reduced Hessian

block can be expressed as:

$$\begin{aligned} (\tilde{M}_k + E_k)^{-1} &= \tilde{M}_k^{-1} + \frac{\lambda_j - \epsilon}{\epsilon \lambda_j} v_j v_j^\top \\ &= \tilde{M}_k^{-1} + \left(\frac{1}{\epsilon} - \frac{1}{\lambda_j}\right) v_j v_j^\top, \end{aligned}$$

and the update of cost-to-go matrix P_k follows as:

$$\begin{aligned} P_k &= \tilde{Q}_k + A_k^\top P_{k+1} A_k - \Gamma_k^\top K_k \\ &= \tilde{Q}_k + A_k^\top P_{k+1} A_k - \Gamma_k^\top \tilde{M}_k^{-1} \Gamma_k - \left(\frac{1}{\epsilon} - \frac{1}{\lambda_j}\right) \Gamma_k^\top v_j v_j^\top \Gamma_k \\ &= P_k^{\text{unmod}} - \left(\frac{1}{\epsilon} - \frac{1}{\lambda_j}\right) \Gamma_k^\top v_j v_j^\top \Gamma_k, \end{aligned} \tag{2.2}$$

Note that this negative rank-1 modification can already render P_k itself indefinite, independently of its downstream effect on $\tilde{M}_{k-1}$, whenever the perturbation magnitude exceeds the smallest eigenvalue of P_k^{unmod} . This occurs even when the regularization on $\tilde{M}_k$ keeps the coefficient bounded, as long as the negative-curvature direction is strongly coupled through the dynamics.

Consequently, the reduced Hessian block $\tilde{M}_{k-1}$ at the previous stage as:

$$\tilde{M}_{k-1} = \tilde{M}_{k-1}^{\text{unmod}} - \left(\frac{1}{\epsilon} - \frac{1}{\lambda_j}\right) z_j z_j^\top, \tag{2.3}$$

where $z_j = B_{k-1}^\top \Gamma_k^\top v_j$ is a column vector.

And the condition number of the regularized reduced Hessian block $\tilde{M}_{k-1}$ can be approximated as:

$$\kappa(\tilde{M}_{k-1}) \approx \frac{c(\epsilon + |\lambda_j|)}{\epsilon |\lambda_j|}, \tag{2.4}$$

where $c = \frac{\|z_j\|^2}{\lambda_{\min}(\tilde{M}_{k-1}^{\text{unmod}})}$ as an $O(1)$ constant.

Notably, when the ϵ is a small positive value, the presence of the negative rank-1 update $-\frac{1}{\epsilon} z_j z_j^\top$ will dominate, not only severely deteriorates its condition number, but also likely to make $\tilde{M}_{k-1}$ indefinite, which can further cause a domino effect through the stages.

2.3.2 Insights and Proposed Methods

This issue reveals two insights in stage regularization:

- The minimum modification objective [7] is not the foremost consideration for stage regularization, as it can cause severe condition number issue in the preceding stages. A trade-off between perturbation norm and condition number must be carefully treated.
- The cost-to-go matrix P_k must be incorporated into the regularization design, as the severity of propagation depends on the coupling structure between stages, which a purely local scheme cannot capture.

2.4 Local Quadratic Convergence Rate for Quadratic Programs

In this section, we will attempt to prove the local quadratic convergence of the regularized interior point method for solving indefinite QPs. Following similar proof structure as in [8], we will organize the proof based on the formulation ?? and start with several assumptions,

Assumption 1. (y^*, λ^*) is a local (or global) minimizer for 1.1.

Assumption 2. Each local minimizer (y^*, λ^*) for 1.1 is isolated.

Assumption 3. Linear independence constraint qualification (LICQ) holds at the minimizer (y^*, λ^*) .

Assumption 4. Strict complementarity holds at the minimizer (y^*, λ^*) .

The Assumptions 1-3 imply the second order sufficient condition (SOSC) for local optimality, so we have the following lemma 1:

Lemma 1. Under Assumptions 1-3, for all $v \in C(y^*, \lambda^*) \setminus \{0\}$, it holds that $v^\top H v > 0$.

Proof. With Assumption 1, The second order necessary condition (SONC) for local optimality holds, which states that for all $v \in C(y^*, \lambda^*)$, we have $v^\top H v \geq 0$.

We proceed by contradiction. Suppose there exists $v \in C(y^*, \lambda^*) \setminus \{0\}$ such that $v^\top H v = 0$. Consider the point $y(\alpha) = y^* + \alpha v$ for $\alpha > 0$. Because the constraints are linear and v is in the critical cone, $y(\alpha)$ remains feasible for sufficiently small α .

Since the objective function of 1.1 is quadratic. Expanding $f(y(\alpha))$ yields:

$$f(y(\alpha)) = f(y^*) + \alpha \nabla f(y^*)^\top v + \frac{1}{2} \alpha^2 v^\top H v$$

As $\nabla f(y^*)^\top v = 0$, given the assumption that $v^\top H v = 0$, the equation reduces exactly to $f(y(\alpha)) = f(y^*)$. This implies that $y(\alpha)$ is also a local minimizer, which directly contradicts Assumption 2. Therefore, $v^\top H v > 0$ must hold. $\square$

Lemma 2. *Under Assumption 1-4, given the sequence $\{W^i\}$ constructed by the algorithm, as $\{(y^i, \lambda^i)\} \rightarrow (y^*, \lambda^*)$, $W^i = \tilde{M}^i = Z^\top \tilde{H}^i Z \succ 0$ for all i sufficiently large.*

Proof. Under Assumption 4, the critical cone $C(y^*, \lambda^*)$ reduces to the tangent space of the active constraints. Therefore, we have $C(y^*, \lambda^*) = \{v \mid Fv = 0, G_{\mathcal{A}}v = 0\}$.

We then proceed with the reduced Hessian. For any $v \in \{v \mid \|v\| = 1\}$, we have:

$$\begin{aligned} v^\top \tilde{M}^i v &= d^\top (H + G^\top T_i^{-1} \Lambda_i G) d \\ &= d^\top H d + \sum_j \frac{\lambda_j}{s_j} (G_j d)^2 \\ &= d^\top H d + \sum_{j \in \mathcal{A}} \frac{\lambda_j}{s_j} (G_j d)^2 + \sum_{j \notin \mathcal{A}} \frac{\lambda_j}{s_j} (G_j d)^2 \end{aligned}$$

where $d = Zv \in \{d \mid Ad = 0\} \setminus \{0\}$.

We now proceed by contradiction, Assume $\exists v_i \in \{v \mid \|v\| = 1\}, v_i \rightarrow \bar{v} : v_i^\top \tilde{M}^i v_i \leq 0$. We can distinguish two cases:

For $G_{\mathcal{A}} \bar{d} = 0$, this indicates $d_i \in C(y^*, \lambda^*) \setminus \{0\}$. $d_i^\top H d_i > 0$ follows by Lemma 1. Rest of the terms are non-negative, so we have $\bar{v}^\top \tilde{M}^i \bar{v} > 0$, contradict with assumption.

For $G_{\mathcal{A}} \bar{d} \neq 0$, the term $\sum_{j \in \mathcal{A}} \frac{\lambda_j}{s_j} (G_j d)^2 \rightarrow \infty$, which dominates the other two terms, and thus $v_i^\top \tilde{M}^i v_i \rightarrow \infty > 0$, contradict with assumption.

Therefore, we can conclude that $W^i = \tilde{M}^i = Z^\top \tilde{H}^i Z \succ 0$ for all i sufficiently large, i.e., $E_i = 0$. $\square$

We take part of the following proposition from [9] as a key step for proof.

Proposition 3. *Let $F : \mathcal{R}^n \rightarrow \mathcal{R}^n$ be twice continuously differentiable and let $w^* \in \mathcal{R}^n$. Let $\rho > 0$ be such that $F(w^*) = 0$ and $\frac{\partial F}{\partial w}(w)$ is invertible whenever*

$w \in \bar{B}(w^*, \rho)$. Let $\delta^N : \bar{B}(w^*, \rho) \rightarrow \mathcal{R}^n$ be the Newton increment $\delta^N(w) := -\left(\frac{\partial F}{\partial w}(w)\right)^{-1} F(w)$. Then given any $c_1 > 0$ there exist $c_2 > 0$ such that the following statement holds:

For every $w \in \bar{B}(w^*, \rho)$ and every $w^+ \in \mathcal{R}^n$ satisfying

$$\|w^+ - (w + \delta^N(w))\| \leq c_1 \|\delta^N(w)\|^2, \quad (2.5)$$

it holds that

$$\|w^+ - w^*\| \leq c_2 \|w - w^*\|^2. \quad (2.6)$$

With the Proposition 3, we will define $w = (y, \lambda)$ as the primal-dual variable and $w^* = (y^*, \lambda^*)$ as the solution, and establish the local quadratic convergence by showing that the conditions in Proposition 3 hold for the iterates generated by the algorithm.

Lemma 4. (Nocedal). Under Assumption 4, when $\{(y^i, \lambda^i)\} \rightarrow (y^*, \lambda^*)$, $\sigma_i \mu_i = O(\|\delta^N\|^4)$.

Proof. By Assumption 4, we have:

$$\mu = \frac{1}{m} \sum_i t_i \lambda_i = O(\|w - w^*\|) = O(\|\delta^N\|).$$

Since the affine-scaling step is the Newton step for $F(w) = 0$, we obtain:

$$\begin{aligned} t_i + \Delta t_i^{\text{aff}} &= t_i^* + O(\|w - w^*\|^2), \\ \lambda_i + \Delta \lambda_i^{\text{aff}} &= \lambda_i^* + O(\|w - w^*\|^2). \end{aligned}$$

It follows that $(t_i + \Delta t_i^{\text{aff}})(\lambda_i + \Delta \lambda_i^{\text{aff}}) = O(\|w - w^*\|^2)$. Thus, $\mu^{\text{aff}} = O(\|w - w^*\|^2) = O(\mu^2)$.

According to the update rule, we obtain $\sigma = (\mu^{\text{aff}}/\mu)^3 = O(\mu^3) \rightarrow 0$. Therefore, combining the orders derived above yields:

$$\sigma \mu = O(\mu^4) = O(\|\delta^N\|^4).$$

□

Lemma 5. Under Assumption 4, when $\{(y^i, \lambda^i)\} \rightarrow (y^*, \lambda^*)$, the affine step size of primal step and dual step satisfy

$$|1 - \alpha_p| = O(\|\delta^N\|), \quad |1 - \alpha_d| = O(\|\delta^N\|).$$

Proof. For the affine step, the linearized complementary condition is given by:

$$t_i \Delta \lambda_i^{\text{aff}} + \lambda_i \Delta t_i^{\text{aff}} = -t_i \lambda_i$$

Dividing both sides by $t_i \lambda_i$ yields the fundamental step identity:

$$\frac{\Delta \lambda_i^{\text{aff}}}{\lambda_i} + \frac{\Delta t_i^{\text{aff}}}{t_i} = -1,$$

where the magnitude bounds for the affine step are given from Lemma 4 as:

$$\|\Delta t^{\text{aff}}\| = O(\|\delta^N\|), \quad \|\Delta \lambda^{\text{aff}}\| = O(\|\delta^N\|).$$

Primal step. As the FTB for primal variables is defined as:

$$\alpha_p^{\max} = \min \left(1, \min_{\Delta t_j < 0} \frac{t_j}{-\Delta t_j} \right),$$

where based on the identity, the ratio $\frac{t_j}{-\Delta t_j}$ can be expressed as:

$$\frac{t_j}{-\Delta t_j} = \frac{1}{1 + \frac{\Delta \lambda_i}{\lambda_i}}.$$

For inactive constraints $t_j \rightarrow t^* > 0$, thus the ratio $\frac{t_j}{-\Delta t_j} \rightarrow \infty$, and such indices do not attain the minimum.

Since $\alpha_p^{\max} \leq 1$, for active constraints, $\lambda_i \rightarrow \lambda_i^* > 0$ and $\Delta \lambda_i = O(\|\delta^N\|)$, we have:

$$1 - \alpha_p^{\max} = 1 - \frac{t_j}{-\Delta t_j} = \frac{\Delta \lambda}{\lambda + \Delta \lambda} = O(\|\delta^N\|).$$

Considering the safeguard value:

$$\tau(\alpha_p^{\max}) = (1 - \alpha_p^{\max})\epsilon_\alpha + (1 - \mu_c)\alpha_p^{\max},$$

where $\mu_c = \min(\epsilon_\mu, \mu)$.

We have:

$$\begin{aligned} \alpha_p &= \tau(\alpha_p^{\max})\alpha_p^{\max} \\ &= \alpha_p^{\max} [1 - (1 - \alpha_p^{\max})(1 - \epsilon_\alpha) - \mu_c \alpha_p^{\max}] \\ &= \alpha_p^{\max} - (1 - \alpha_p^{\max})(1 - \epsilon_\alpha)\alpha_p^{\max} - \mu_c (\alpha_p^{\max})^2 \end{aligned}$$

In local convergence, $\mu_c = \mu$ holds, by lemma 4, the $1 - \alpha_p$ is :

$$\begin{aligned} 1 - \alpha_p &= O(\|\delta^N\|) + O(\|\delta^N\|) + O(\|\delta^N\|^4) \\ &= O(\|\delta^N\|). \end{aligned}$$

Dual step. Symmetrically, the FTB for dual variables is defined as:

$$\alpha_d^{\max} = \min \left(1, \min_{\Delta\lambda_j < 0} \frac{\lambda_j}{-\Delta\lambda_j} \right),$$

For active constraints, corresponding indices do not attain the minimum, due to $\frac{\lambda}{-\Delta\lambda} \rightarrow \infty$. For inactive constraints, $t_j \rightarrow t_j^* > 0$ and $\Delta t_j = O(\|\delta^N\|)$, thus the ratio $\frac{\lambda_j}{-\Delta\lambda_j} = 1 - O(\|\delta^N\|)$.

We then have:

$$1 - \alpha_d^{\max} = 1 - \frac{\lambda_j}{-\Delta\lambda_j} = O(\|\delta^N\|).$$

Following the same logic as primal step, we can conclude that $1 - \alpha_d = O(\|\delta^N\|)$.

Combining the two estimates gives

$$|1 - \alpha_p| = O(\|\delta^N\|), \quad |1 - \alpha_d| = O(\|\delta^N\|).$$

□

Theorem 6. *Under Assumption 1-4, Let $\{y^i\}$, $\{\lambda^i\}$ as sequences constructed by algorithm, $\{(y^i, \lambda^i)\}$ converges to the solution (y^*, λ^*) with local quadratic convergence rate.*

Proof. We establish that the conditions in Proposition 3 hold with ?? as:

$$\underbrace{\begin{bmatrix} M + E^i & Z^\top G^\top \\ GZ & -\Lambda_i^{-1} T_i \end{bmatrix}}_{\mathcal{M}(w)} \begin{bmatrix} \Delta y_z \\ \Delta \lambda \end{bmatrix} = - \begin{bmatrix} r_{\text{stat, redu}}^i \\ \tilde{r}_{\text{ineq, redu}}^i \end{bmatrix}, \quad (2.7)$$

Under Assumptions 1-4, $\frac{\partial \mathcal{M}}{\partial w}$ is invertible at w^* . By Lemma 2, $E^i = 0$ for all i sufficiently large, so the left-hand side coincides with the exact Newton Jacobian. By Lemma 4, $\sigma_i \mu_i = O(\|\delta^N\|^4)$. Consequently, $\|\Delta w_i - \delta^N(w_i)\| = O(\|\delta^N(w_i)\|^4)$ for $w^i \rightarrow w^*$.

With $\Delta w_i = [\Delta y_i, \Delta \lambda_i]^\top$, we start with the update of primal variables y ,

$$\|y_{i+1} - (y_i + \delta_y^N)\| = \|y_i + \alpha_p \Delta y_i - (y_i + \delta_y^N)\| \quad (2.8)$$

$$\leq |1 - \alpha_p| \|\Delta y_i\| + \|\Delta y_i - \delta_y^N\| \quad (2.9)$$

$$\leq C_1 \|\delta_y^N\| \|\Delta y_i\| + C_2 \|\delta_y^N\|^4 \quad (2.10)$$

$$\leq C' \|\delta_y^N\|^2, \quad (2.11)$$

As a result, the condition 2.5 in Proposition 3 holds for the primal variables. For the dual variables λ , the same logic applies with α_d substituting α_p .¹

Thus, we can conclude that the sequence $\{(y^i, \lambda^i)\}$ converges to (y^*, λ^*) with local quadratic convergence rate. $\square$

¹In the implementation, the update for the dual variables is taken as $\max(\lambda + \alpha_d \Delta \lambda, \epsilon)$, where ϵ is on the order of machine precision. This truncation is only activated when the accuracy reaches $\mathcal{O}(\sqrt{\epsilon})$ and thus does not affect the asymptotic convergence rate analysis above.

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